

Single-Cell RNA Sequencing Analysis of Leukemia versus Normal Hematopoietic Cells

A Comparative Evaluation of Three CellTypist Annotation Models for Differential Gene Expression



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Single-Cell RNA-Seq Analysis Report

Abstract

This report presents a single-cell RNA sequencing (scRNA-seq) analysis comparing a Leukemia sample (971 cells) and a Normal sample (760 cells), sequenced on the Illumina HiSeq 4000 platform. After quality control, normalization, highly variable gene selection, PCA, and Leiden clustering, cells were annotated into biological cell types using three independent CellTypist reference models. Differential gene expression (DGE) between Leukemia and Normal cells was then evaluated within each annotated cell type (Wilcoxon rank-sum test; $|\log_2FC| \geq 1$, adjusted $p < 0.05$). The hematopoietic stem/progenitor (HSC-MPP) population was the only cluster testable under all three models. Model 2 identified 48 significant differentially expressed genes (DEGs) in this population, versus 4 each for Models 1 and 3, indicating that Model 2 provided the most informative and statistically robust description of differential expression in this dataset.

Objective

- Process and quality-control raw single-cell expression data from Leukemia and Normal samples.
- Cluster cells and annotate cell types using three independent CellTypist reference models.
- Identify differentially expressed genes between Leukemia and Normal cells within each annotated cell type.
- Compare the three annotation models and determine which best describes the differential gene expression between conditions.

Introduction

Single-cell RNA sequencing enables gene expression profiling at individual-cell resolution, resolving cellular heterogeneity that bulk RNA-seq averages out. This is especially useful in leukemia, where malignant transformation arises within specific hematopoietic stem and progenitor populations while other blood lineages remain comparatively normal. Comparing a Leukemia sample against a healthy Normal reference at single-cell resolution localizes disease-associated transcriptional changes to the specific populations in which they occur.

A key step in this workflow is automated cell type annotation, performed here using CellTypist, which assigns a label to each cell by comparison against a pre-trained reference model. Because reference models differ in granularity and biological scope, the same dataset can be annotated differently depending on the model applied. This report evaluates three CellTypist models side by side and compares the resulting differential expression outcomes, to identify which model most reliably captures the transcriptional differences between Leukemia and Normal cells.

Methodology

Dataset and Quality Control

Two 10x Genomics-format samples were used: Normal (760 cells) and Leukemia (971 cells), each with 36,601 genes. The samples were concatenated (1,731 cells total) with sample identity retained for group comparisons. Per-cell QC metrics (total counts, genes detected, counts-per-gene, %mitochondrial, %ribosomal) were computed, and cells were kept only within the 1st-99th percentile range for counts/genes/complexity, with mitochondrial content $< 15\%$ and ribosomal content $< 40\%$. This reduced the dataset to 323 cells (1,408 cells, 81.4%, removed).

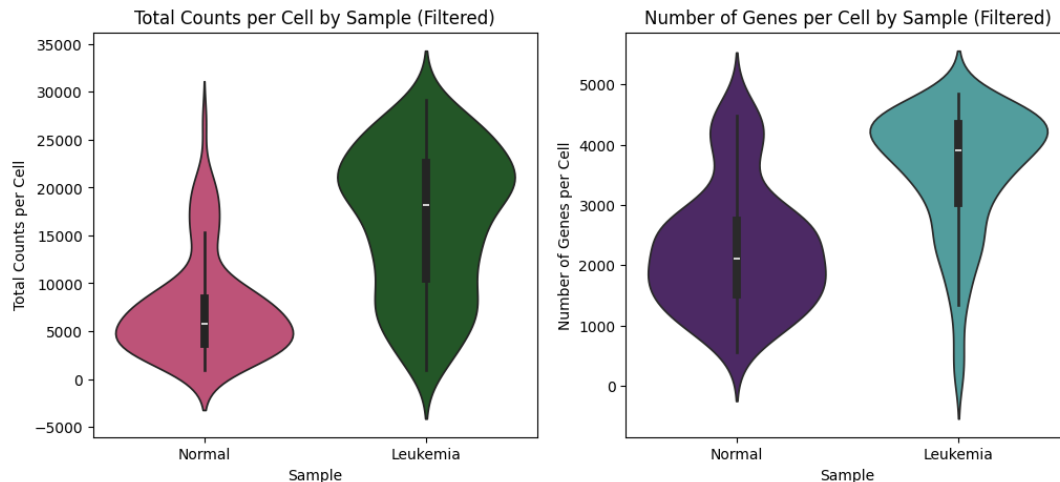


Figure 1. Total counts and gene counts per cell, by sample, after QC filtering.

Normalization, Clustering and Annotation

Filtered counts were library-size normalized (target sum = $1e4$) and $\log_{1p}$ -transformed. The top 2,000 highly variable genes were used for PCA (~50 PCs captured 80% of variance) and neighborhood-graph construction; Leiden clustering (resolution 0.7) produced four clusters (93, 88, 79, 63 cells). Cell type identity was then assigned with CellTypist majority voting, run independently with three reference models: Model 3 (Adult_chSPCs_Illumina.pkl, trained on human HSPCs) produced seven labels (B, CLP, ERY, GMP-L, HSC_MPP, MEBEMP-L, MKP); Model 1 (Immune_All_Low.pkl) produced nine broader hematopoietic/stromal-like labels; Model 2 (Immune_All_High.pkl) produced a coarser annotation in which Erythroid and HSC-MPP were the only populations represented in both conditions with enough cells to test.

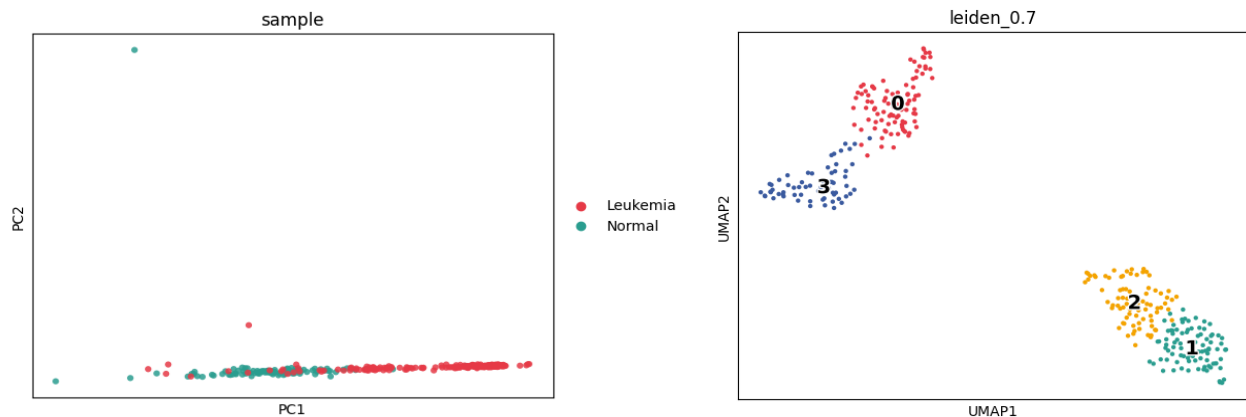


Figure 2. Left: PCA embedding colored by sample. Right: UMAP colored by Leiden cluster (resolution 0.7).

Differential Expression Analysis

For each annotated cell type, under each of the three models, Leukemia cells were compared against Normal cells (reference) using `sc.tl.rank_genes_groups` (Wilcoxon test, $\log$ -normalized data). Genes were called significant at adjusted $p < 0.05$ and $|\log_2FC| \geq 1$. Cell types with only one condition present, or fewer than three cells in either group, were excluded from testing.

Results

Cell Type Annotation (Model 2)

Model 2 annotated the 323 quality-controlled cells into five broader categories (Table 1). Compared with Model 3's seven-category scheme, Model 2 pools cells into fewer, larger groups; most notably HSC/MPP, which grew from 66 cells (Model 3) to 105 cells (Model 2).

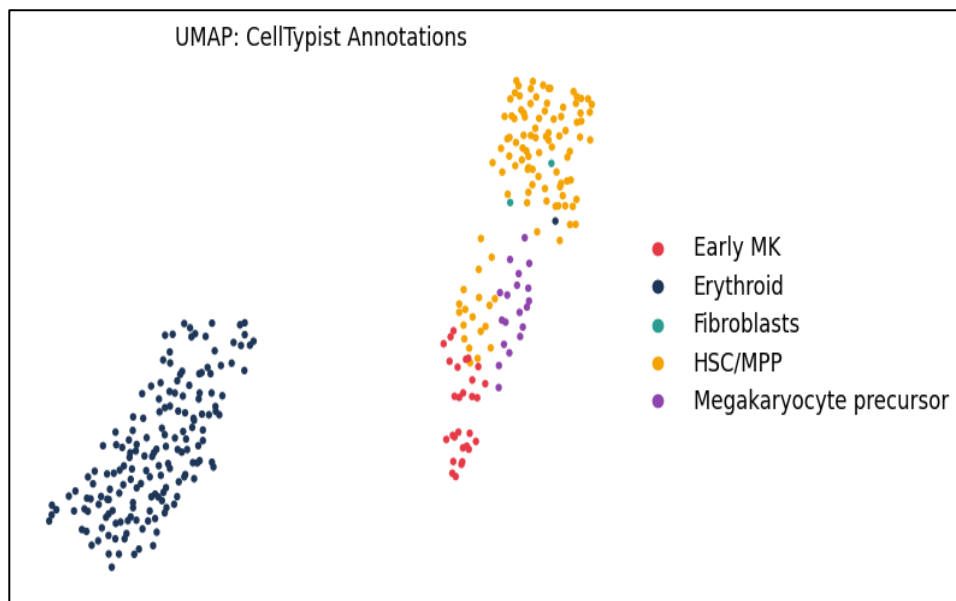


Figure 3. UMAP colored by Model 2 (CellTypist) cell type annotation.

Cell Type(Model 2)	Cell Count	% of Total
Erythroid	168	52.0%
HSC/MPP (Stem / multipotent progenitor)	105	32.5%
Early MK	29	9.0%
Megakaryocyte precursor	19	5.9%
Fibroblasts	2	0.6%

Table 1. CellTypist Model 2 annotation counts across the 323 filtered cells.

Differential Expression Outcomes Across Models

DGE testing was only possible for cell types with at least three cells in both groups; most populations failed this criterion because Normal contributed very few cells to several clusters. Table 2 summarizes every cell type for which testing was actually performed.

Model	Cell Type	Normal (n)	Leukemia (n)	Significant DEGs
Model 1	HSC/MPP	67	13	4
Model 1	Megakaryocyte precursor	20	8	0
Model 1	Mid erythroid	imbalanced*	imbalanced*	0
Model 2	HSC-MPP	imbalanced*	imbalanced*	48
Model 2	Erythroid	imbalanced*	imbalanced*	0
Model 3	HSC_MPP	53	13	4
Model 3	MKP	3	46	0
Model 3	ERYP	3	165	0

Table 2. DGE testing outcome per cell type/model (Wilcoxon, $p_{adj} < 0.05$, $|\log_2 FC| \geq 1$); both groups still met the minimum 3-cell threshold.

Direct Comparison on the HSC-MPP Population

HSC-MPP was the only cell type annotated with sufficient cells for testing under all three models, making it the most reliable basis for comparison (Figure 4).

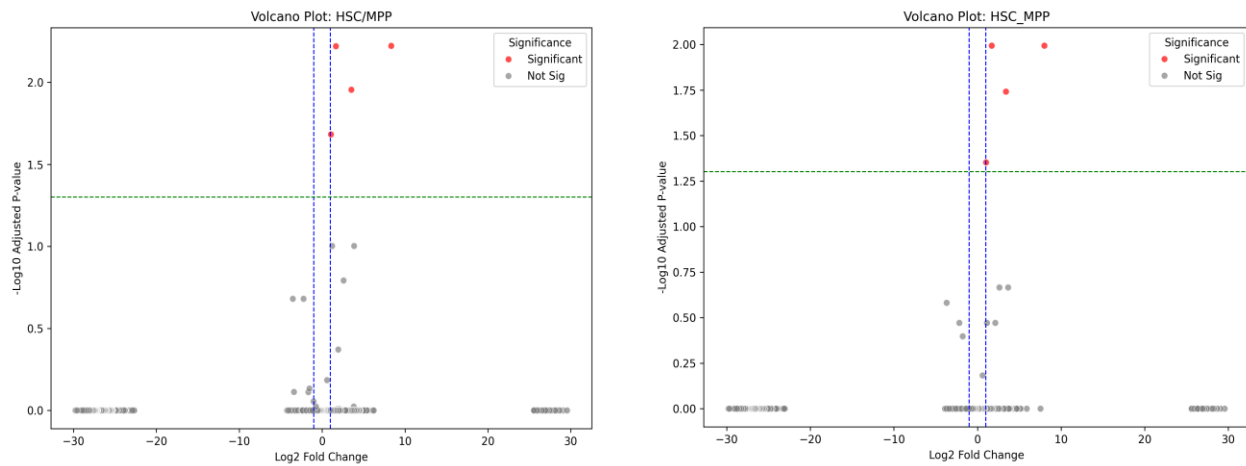
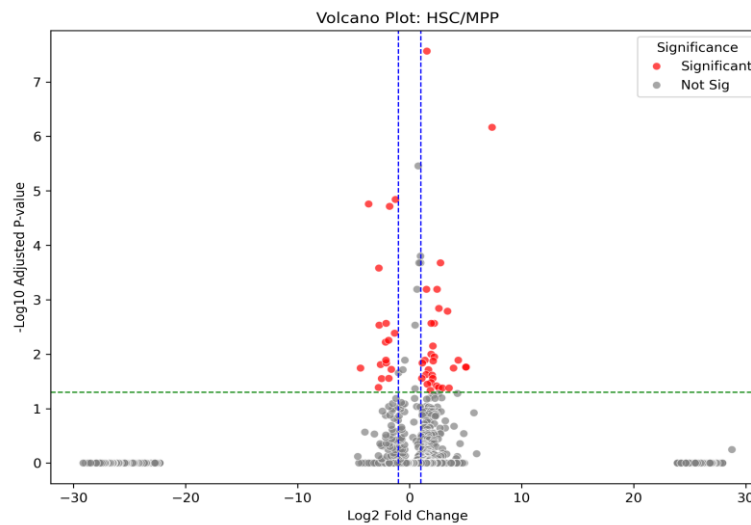


Figure 4. HSC-MPP volcano plots.: Model 1 (4 sig. genes).

Model 3 (4 sig. genes).



Model 2 (48 sig. genes).

Models 1 and 3 converged on the same four significant genes (HBB, RPS26, KCNQ5, RPL39; all up in Leukemia). Model 2 identified a much larger, biologically richer set of 48 significant genes, including hemoglobin genes (HBB, HBD), HLA/immune genes (HLA-DRB5, CD74, CD52, CD53, CD37), cell-cycle genes (TUBA1B, TUBB, CENPP, CENPK, H2AFZ), and stress/immediate-early genes (FOS, JUN, DUSP1). Top genes are listed in Table 3.

Gene	log2 Fold Change	Adjusted p-value	Direction
RPS26	1.54	2.7×10^{-8}	Up in Leukemia
HBB	7.36	6.8×10^{-7}	Up in Leukemia
HLA-DRB5	-3.66	1.7×10^{-5}	Down in Leukemia
CD74	-1.78	1.9×10^{-5}	Down in Leukemia
DUSP1	-2.74	2.6×10^{-4}	Down in Leukemia

FOS	-2.14	6.0×10^{-3}	Down in Leukemia
JUN	-2.11	1.3×10^{-2}	Down in Leukemia

Table 3. Top significant DEGs identified by Model 2 in HSC-MPP (full 48-gene list in the accompanying CSV).

Discussion: Which Model Best Describes the DEGs

Model 2 best describes the differentially expressed genes between Leukemia and Normal cells. On the one population tested under all three models (HSC-MPP), Model 2 detected twelve times more significant genes (48) than Model 1 or Model 3 (4 each), and its genes are biologically coherent: down-regulation of antigen-presentation/immune-activation genes (HLA-DRB5, CD74, CD52, CD53, CD37) alongside up-regulation of proliferation/cell-cycle genes (TUBA1B, TUBB, CENPP, CENPK) is consistent with the loss of normal immune marker expression and increased proliferative activity that characterizes leukemic stem/progenitor cells. Models 1 and 3 converged on the same narrow set of four genes, mostly ribosomal (RPS26, RPL39) and less specific to leukemia biology, suggesting Model 2's grouping better balanced statistical power against biological specificity.

For every other annotated cell type, all three models returned zero significant genes. This reflects sample size rather than model failure: after QC filtering, only 323 of 1,731 cells remained, and Normal contributed as few as 2-5 cells to several clusters (B, CLP, GMP-L), leaving too little power for the Wilcoxon test regardless of annotation model.

Output of model2 DEGs:

```
Processing cell type: Megakaryocyte precursor
sample
```

```
Normal      17
```

```
Leukemia    2
```

```
Name: count, dtype: int64
```

```
Skipping Megakaryocyte precursor: one or more groups has too few cells (<3)
```

```
Processing cell type: HSC/MPP
```

```
sample
```

```
Normal      75
```

```
Leukemia    30
```

```
Name: count, dtype: int64
```

```
DEGs saved for HSC/MPP: total=36601, significant=48
```

```
Volcano plot saved for HSC-MPP
```

```
Processing cell type: Fibroblasts
```

```
sample
```

```
Normal      2
```

```
Name: count, dtype: int64
```

```
Skipping Fibroblasts: only one sample group present, cannot run DE
```

```
Processing cell type: Erythroid
```

```
sample
```

```
Leukemia    165
```

```
Normal      3
```

```
Name: count, dtype: int64
```

```
DEGs saved for Erythroid: total=36601, significant=0
```

```
Volcano plot saved for Erythroid
```

```
Processing cell type: Early MK
```

```
sample
```

```
Leukemia    29
```

Name: count, dtype: int64

Skipping Early MK: only one sample group present, cannot run DE

Conclusion

This analysis processed and quality-controlled a Leukemia versus Normal single-cell dataset, clustered the cells, and annotated them using three independent CellTypist models. Differential expression testing was statistically meaningful only for the HSC-MPP population, adequately represented in Leukemia and Normal groups under all three models. Model 2 produced the most statistically robust and biologically interpretable DEG set (48 significant genes, including immune and cell-cycle markers) and is recommended as the most suitable annotation model for describing differential expression in this dataset. Increasing the Normal sample size in future work would substantially improve statistical power across the remaining cell types.